Data Patterns and Cluster Profiles

The cluster analysis in this module activity shows that higher education programs naturally separate into distinct financial groups. These groups are based on student debt, enrollment size, and long-term earnings. The largest group is the baseline profile, which contains 557 out of the 1,000 sampled programs. This group represents typical academic tracks with stable enrollment numbers and steady wage growth over five years. In contrast, the high-value group highlights a selection of high-performing programs where students get a strong return on investment. In these programs, median earnings grow sharply from Year 1 to Year 5, while average student debt remains low. Finally, the remaining segments isolate high-debt programs where student debt loads are heavy but early-career salaries stay low. These empirical types of patterns prove that a program's specific cluster profile is a much better predictor of a student's financial future than the overall size or enrollment of the school.

Visual Validation of Findings

The charts validate these findings by making complex numeric relationships easy to see without forcing the reader to parse raw data tables. Choosing the proper chart layout is essential for ensuring that data values accurately communicate patterns to an audience without causing confusion (insightsoftware, n.d.). For example, the PCA scatterplot simplifies seven different variables into a flat, two-dimensional view. The four groups show clear physical separation on the plot, and the primary axis accounts for 53.39% of the data variance. This clear separation proves that the clustering algorithm successfully completed the core exploratory data analysis step of identifying real structural groups rather than random data patterns (ZyBooks, 2026) This group behavior is further confirmed over time by the trend chart, which tracks median earnings at the 1-year, 4-year, and 5-year milestones. The lines for each cluster run parallel and do not cross. This layout proves that the income gap between high-performing programs and high-debt programs steadily widens over five years. Building these advanced visual layouts smoothly was made possible by using the Seaborn library. Because Seaborn is built directly on top of Matplotlib, it provides a simple interface that streamlines statistical graphics and integrates naturally with data frames (Turp 2023; ZyBooks, 2026). This integration allowed for the deliberate use of distinct colors and circular markers on the lines to make the changing financial trajectories immediately clear. Lastly, the cluster size bar chart counts how many programs fall into each group. The vertical bars give quick context, showing that most programs belong to the stable middle tier, while the high-performing and high-debt groups are smaller statistical outliers (ZyBooks, 2026).

References

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